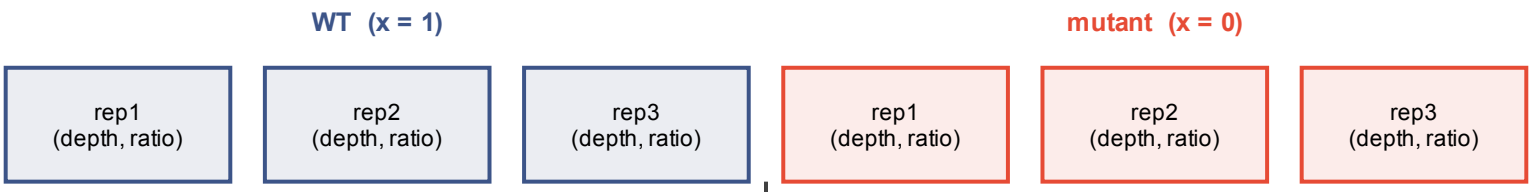
How call_dependent_methylKit.R tests one candidate site

1 | Input — one candidate site, six libraries



each sample → counts: modified $k = \text{round}(\text{depth} \times \text{ratio})$, unmodified $\text{depth} - k$
depth matters: 10/20 carries more weight than 1/2

depth = 0 replicate → treated as MISSING (not "modification lost")
excluded from Mut_Ratio mean and from the test: 3v3 → 3v2
if a whole group is wiped out → p = NA

2 | Weighted binomial logistic regression (per site)

$\text{logit}(\pi) = \beta_0 + \beta_1 \cdot x$ x = 1 (WT) / 0 (mutant), weight = depth
$\beta_1 > 0$ means WT > mutant → the dependent direction

3 | Overdispersion-corrected test

overdispersion factor φ (MN)

$\varphi > 1$ → F test
deviance/$\varphi \sim F(1, n-2)$
penalizes noisy replicates

$\varphi \leq 1$ → χ^2 test
ordinary likelihood ratio
replicates consistent

two-sided p-value
→ BH q-value across all candidate sites

4 | Direction + FDR

meth.diff > 0 (WT higher) AND $q < 0.05$ → passes
wrong direction or $q \geq 0.05$ → never passes

5 | Decision gates — all four must hold

- mock mean depth ≥ 10 — site well measured in mutant mock
- BH $q < 0.05$ — significant loss in mutant (FDR)
- $\Delta = \text{WT_Ratio} - \text{Mut_Ratio} \geq 0.05$ — effect-size floor
- both mutant lines pass all above — line 1 AND line 2

Dependent site